PROJECT REPORT

STUDENT STRESS LEVEL PREDICTOR

1. INTRODUCTION

In today's academic environment, students often experience high levels of stress due to continuous academic pressure,

deadlines, and lack of proper rest. Stress may cause negatice impact on both mental and physical health. Therefore, identifying

stress at an early stage is important. This project aims to use machine learning techniques to predict student stress levels

based on simple lifestyle factors.

2. PROBLEM STATEMENT

The objective of this project is to develop a machine learning model that can predict whether a student is stressed or not using

the following inputs:

- Study hours
- Sleep hours

The model classifies stress into two categories:

- 0 → Low Stress
- 1 → High Stress

3. MOTIVATION

This problem was chosen because stress among students is very common and often ignored. A simple predictive system can help in early

detection and encourage better lifestyle habits. This project demonstrates how machine learning can be applied to solve everyday problems.

4. DATASET

A dataset was manually created for this project due to time constraints. It includes:

- Study Hours: Number of hours spent studying per day
- Sleep Hours: Number of hours of sleep per day
- Stress Level: Target variable indicating stress (0 or 1)

Although the dataset is small, it is sufficient to demonstrate the working of a machine learning model.

5. METHODOLOGY

The following steps were followed in this project:

- Data creation using pandas DataFrame
- Selection of relevant features (study hours and sleep hours)
- Splitting the dataset into training and testing sets
- Training the model using Logistic Regression
- Evaluating the model using accuracy score
- Making predictions based on new input values

6. MODEL USED

Logistic Regression was chosen for this project as it is a simple and effective algorithm for binary classification problems. It helps in

predicting the probability of a class and is easy to implement.

7. RESULTS AND OBSERVATIONS

The model was successfully trained and tested. It was able to predict stress levels based on given inputs.

Example:

Study Hours = 6, Sleep Hours = 4

Prediction = High Stress

The accuracy of the model was reasonable considering the small dataset size. It shows that both study hours and sleep hours have an impact on stress levels.

8. CHALLENGES FACED

- Difficulty in installing and setting up required Python libraries
- Limited dataset size
- Initial confusion in understanding machine learning workflow

- These challenges were resolved through practice and debugging.

9. CONCLUSION

This project successfully demonstrates the use of machine learning in solving a real-world problem. It provides a simple approach to predicting stress

levels using basic input features. The project helped in understanding key concepts such as classification, model training, and prediction.

10. LEARNING OUTCOMES

- Learned how to implement a machine learning model
- Understood the importance of data preprocessing
- Gained hands-on experience with Python libraries like pandas and scikit-learn
- Learned how to debug errors and run code successfully

11. REAL-WORLD APPLICATIONS

This model can be used in:

- Student counseling systems
- Mental health monitoring tools
- Educational institutions to track student well-being

12. FUTURE SCOPE

- Use larger and real-world datasets
- Include more features such as screen time, physical activity, and social interaction
- Improve model accuracy using advanced algorithms
- Develop a graphical user interface for better usability

13. PERSONAL REFLECTION

This project was implemented by me as part of my coursework. Initially, I faced issues while setting up the environment

and running the code, but resolving these helped me understand the workflow better. This project improved my confidence in working with machine learning concepts.